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SSL-XIoMT: Secure, Scalable, and Lightweight Cross-Domain IoMT Sharing With SSI and ZKP Authentication

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ABSTRACT The Internet of Medical Things (IoMT) is transforming healthcare by enabling devices to generate and share critical patient data. However, securely sharing this data across different healthcare entities remains a significant challenge due to concerns over privacy and security. Traditional solutions using Ciphertext Policy Attribute-Based Encryption (CP-ABE), Self-Sovereign Identity (SSI), and Zero-Knowledge Proofs (ZKPs) offer secure and anonymous data access, but they often fall short in scalability and integration, particularly in cross domain environments. To address these limitations, we introduce SSL-XIoMT, an optimized SSI and ZKP authentication framework within a consortium Hyperledger-based environment. This innovative system integrates SSI under advanced Zero-Knowledge Scalable Transparent Argument of Knowledge (ZK-STARK) and Plonk protocols within a consortium Hyperledger framework for privacy-preserving identity verification. We enhance identity privacy by integrating Multi-Party Computation (MPC), ensuring that identity credentials and ZKP proofs are securely shared and reconstructed without exposing sensitive information. Additionally, we optimize CP-ABE by offloading complex computations to fog nodes, which pre-compute attributes and logical operations. This approach significantly reduces computational overhead and enhances both privacy and efficiency. Our extensive analysis shows that SSL-XIoMT dramatically improves the performance of processing time for CP-ABE encryption and decryption compared to current methods. Moreover, our hybrid ZKPs based authentication approach outperforms the existing schemes regarding processing time and flexibility. The throughput test also demonstrates that SSL-XIoMT is practical for large scale cross-domain data sharing implementation.

INDEX TERMS CP-ABE, hyperledger, IoMT, merkle-tree, fog nodes, multi-party computation, SSI, plonk, ZK-STARK.

I. INTRODUCTION

The Internet of Medical Things (IoMT) is revolutionizing the healthcare sector by generating and processing vital healthcare data through smart devices. Security awareness is the most essential part to ensure the safeness of the system from any misbehave acts while maintaining the performance scalability is also necessary such as the fine-grained access control for offloading cryptographic operations, as seen in several works such as [1], [2], [3], [32], [33], [43], [44], [45]. Regarding security, authentication is the initial stage to be leveraged. Recently, there are some works [4], [5], [6], [7], [8], [9] that introduced cross-domain authentication with the integration of blockchain technology in a combination of Elliptic curve cryptography, certificate, and identity-based authentication.Although they were commonly used, implementing ECC with blockchain can be complex and requires careful management of cryptographic keys. As for certificate-based authentication, they have limitations on identity revocation that need to be improved to achieve scalability and lower the centralization without the need of trusted certificates authorities all the time.

To address these challenges, identity-based blockchain authentication has gained popularity due to its strengths in providing secure, decentralized data management while simplifying authentication processes, which is crucial for resource-constrained IoMT devices. Self-Sovereign Identity

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(SSI) is one of the most significant identity-based blockchain management systems that has been utilized in many schemes, such as [10], [11], [12], [13], [14], [15], [16]. SSI embodies an authentication protocol that grants users comprehensive control over their identities, enhancing verification processes, mitigating data leaks, and simplifying access.

Zero-Knowledge Proof (ZKP), as discussed in [17], [18], [19], is a mechanism designed to protect identities during cross-domain authentication and authorization checks before users are allowed to perform certain operations. ZK-SNARK (Zero-Knowledge Succinct Non-Interactive Argument of Knowledge) is one of the most popular and widely adopted protocols for anonymous authentication in distributed systems. It has demonstrated fast proof generation and verification. However, it has notable limitations, including the need for a trusted setup, lack of post-quantum resistance, and higher long-term setup costs.

To address these drawbacks, ZK-STARK (Zero-Knowledge Scalable Transparent Argument of Knowledge) was introduced in 2018, as found in [20]. ZK-STARK is designed to overcome the limitations of ZK-SNARK by eliminating the requirement for a trusted setup, offering resistance to post-quantum attacks, and supporting anonymous authentication.

In large-scale, dynamic, and cross-domain environments such as healthcare data sharing, the need for privacy-preserving, efficient, and adaptive authentication is paramount. Therefore, relying solely on ZK-SNARK or ZK-STARK is not ideal, as it necessitates compromising either security or performance. In 2019, A. Gabizon et al. [21] introduced PLONK, a variation of ZK-SNARK that serves as a middle ground between ZK-SNARK and ZK-STARK. PLONK requires only a universal setup for proof generation and verification, offering some improvements over traditional ZK-SNARKs. However, it still faces challenges such as higher cryptographic complexity and lacks support for post-quantum resistance compared to ZK-STARK. Furthermore, while ZKP provides anonymous authentication, it alone is insufficient if a verifier is compromised. When multiple stakeholders (e.g., hospitals, IoMT nodes, fog nodes) need to collaboratively process private data.

Moreover, the cryptographic protocol used to support secure sharing among users in different domains is also non-trivial. Ciphertext Policy Attribute-Based Encryption (CP-ABE) has been intensively adopted as a suitable solution for supporting fine-grained and secure outsourced data. However, CP-ABE has major drawbacks related to the high computation cost associated with pairing and exponentiation, especially when the access policies used to encrypt the data consist of many attributes. Therefore, applying CP-ABE in an IoT environment is not suitable. To date, numerous recent studies [1], [2], [3], [22], [23], [43], [44] have proposed CP-ABE outsourcing methods that offload full or partial encryption and decryption to fog nodes or proxies to reduce the computational burden on the data owner's side. However, this may result in delays when processing such expensive operations at the proxy or a particular fog node, especially when there is a huge volume of CP-ABE encryption or decryption requests, leading to overall system degradation.

Furthermore, applying CP-ABE in cross-domain data sharing by engaging multi-authority systems [24], [25], [26] introduces high complexity for key management and high dependency on multiple attribute authorities.

In this article, we tackle all the above challenges related to efficient, secure, and flexible ZKP authentication and optimized CP-ABE setting. We introduce our new access control scheme called SSL-XIoMT to deliver secure, scalable, and lightweight access control for cross-domain healthcare IoMT data sharing. The contributions of our proposed system are summarized as follows:

1. We introduced a new design and development of SSI management for cross-domain authentication using DID and integrated the hybrid approach of two popular ZKP protocols including PLONK and ZK-STARK for authorization across multiple domains. We also applied MPC to facilitate the privacy-preserving sharing and construction of identity credentials and ZKP proofs. With this combination, our system can strengthen security against post-quantum attacks, and lightweight authorization check via PLONK protocol.
2. We developed a novel optimization for CP-ABE encryption and decryption by introducing a formalism for pre-computing access policy elements prior to pairing. The goal is to minimize computational overhead at the fog nodes, especially as the number of user requests and interactions rapidly escalates.
3. We devised a PRE protocol designed to achieve both the verification of the authenticity of the requestor and check for data integrity before performing CP-ABE decryption. This protocol selectively re-encrypts the data using public key encryption before sharing it with the requester, eliminating the need to deal with the Multi-Authority (MA) CP-ABE protocol.
4. We integrate Hyperledger technology to support customizable smart contracts which can assist all operations to be fully traceable and automated.

The rest of the article is demonstrated about the related works in the Section II, the proposed system model in Section III, security analysis in Section IV, performance evaluation in Section V, and conclusion in Section VI.

II. RELATED WORK

Cross-domain authentication is increasingly important for secure data sharing across various domains, requiring authentication before other security mechanisms. There are a number of works in [4], [5], [6], [7], [8], [9] where authors proposed techniques for cross-domain authentication and cross-domain data sharing [27]. There are several mechanisms to support cross-domain authentication integrated blockchain such as certificate-free-based, certificate-based, ECC-based, identity-based, and zero knowledge proof-based authentication. In [4], Liu et al. developed an authentication scheme using ECC-Elgamal-based PCE for key negotiation,

encryption, and digital signature verification. Their system processes requests in parallel through a sharding blockchain-based certificate authority (SBCA) across secure domains and applies identity-based access control (IBAC) for policy management. In [5], Hao et al. ensures system authenticity, using a lightweight consortium blockchain for IoT cross-domain access control, with all activities recorded in PBFT consensus. In [6], Dong et al. proved the security of their blockchain-based cross-domain authentication scheme for the industrial internet through SVO logical proof and formal analysis with Scyther, addressing key escrow and certificate management issues. In [7], Liu et al. developed a blockchain-assisted distributed key generation and authentication system, embedding public keys and authentication data to enhance scalability and security in cross-domain IoT environments, while eliminating reliance on centralized authorities. In [8], Cheng et al. combined blockchain, certificateless cryptography, ECC, and pseudonym-based cryptography to support mutual authentication and privacy protection between edge servers and IoT devices.

Among the blockchain-based, identity-management is one of the trending techniques to be used for cross-domain authentication. In [9], Chen et al. introduced a one-way accumulation structure for decentralized identity management, improving authentication efficiency and reducing identity validation time. In scheme [10], [11], [12], [13], [14], [15], [16] SSI is the identity management that serves as the solution for cross-domain authentication. It guarantees the fully controllable identities for a user which are required to be authenticated without exposing any non-necessary information of the user. Several surveys work from scheme [10], [11], [12], [13], [14] have explored the real use cases of SSI, available wallets, architecture, pros and cons, and the open-source implementations available. In scheme [15], Xu et al. proposed a blockchain-based SSI management system for mobile networks, which supports dynamic user revocation using chameleon hash and reduces delay and storage overhead. In scheme [16], Zeydan et al. designed a blockchain-based SSI framework for multi-cloud service management, leveraging Hyperledger Indy for decentralized identity verification and Quorum blockchain for secure transaction logging. This approach ensures privacy preservation and data integrity through DID-based credential management and cryptographic identity verification, while smart contracts facilitate transparent and scalable network service orchestration. In [17], Luong et al. implemented ZK-SNARKs and Shamir's Secret Sharing (SSS) for privacy-preserving identity management. In [18], Dieye et al. enhanced SSI with dual ZKP protocols, enabling identity verification without disclosing personal information, while complying with GDPR and eIDAS. In [19], Ra et al. combined ZK-SNARKs and CL-Signature for secure, anonymous verification in identity-based blockchain systems, with implementation on Hyperledger Fabric demonstrating broad application and strong privacy protection. Due to the increasing reliance on cloud outsourcing for data flexibility and availability, lightweight and secure encryption mechanisms have become essential for organizations, institutions, and enterprises aiming to protect sensitive information. Prior studies [1], [3], [22], [23], [28], [29] have commonly adopted AES in combination with CP-ABE to achieve multi-layered encryption for ensuring data confidentiality and privacy. In [1], Fugkeaw et al. proposed a novel offloading framework for IoT devices, performing partial CP-ABE encryption at an aggregated terminal to reduce computation overhead. Similarly, Wang et al. [27] introduced FABRIC, a cross-domain proxy re-encryption scheme that transforms ciphertexts from identity-based to policy-based encryption. Their design supports an unbounded number of attributes with constant-size public parameters, improving scalability in multi-attribute systems. Several works [2], [3] also addressed the efficiency of CP-ABE by offloading partial encryption and decryption tasks to cloud-based proxies, enabling mobile or resource-limited users to participate with minimal overhead. Fugkeaw et al. [22] further demonstrated the effectiveness of proxy servers in supporting lightweight updates for outsourced personal health records using CP-ABE. Meanwhile, Hu Xiong et al. proposed unbounded and revocable ABE schemes for fine-grained access control in digital twins [28] and secure, searchable data sharing in autonomous transportation systems [29]. In the context of cross-domain sharing, Guo and Idrees et al. [23], [26] presented blockchain-based attribute-based encryption frameworks tailored for 6G-enabled VANETs, incorporating multi-tree revocation and batch joining mechanisms. Despite these advances, applying CP-ABE directly to cross-domain environments remains non-trivial. Multi-Authority CP-ABE (MA-CP-ABE), though designed for decentralized control, introduces significant computational and management overhead, making it unsuitable for resource-constrained or latency-sensitive settings like IoMT. To the best of our knowledge, no existing work has comprehensively addressed lightweight, privacy-preserving, and anonymous cross-domain authentication and data sharing in the IoMT domain, especially with real-time verifiability and ZKP support.

III. OUR PROPOSED SCHEME

In this section, we present the system model and cryptographic contrutctions of our proposed SSL-XIoMT scheme. Fig. 1 shows SSL-XIoMT system model.

A. SYSTEM MODEL

We proposed a secure and self-sovereign identity based blockchain cross-domain authentication with ZKP authorization IoMT data sharing with the offloading partial CP-ABE encryption and decryption. Fig. 1 illustrates the system overview of our proposed scheme.

The system model consists of the following entities:

1) Data Owner (DO): refers to the patients whose health information has been collected by IoT devices. DO sets the policy specifying who can access their personal health record.
2) Data Users (Dus): consists of both Data Owners and Data Users who make a query to obtain the IoMT

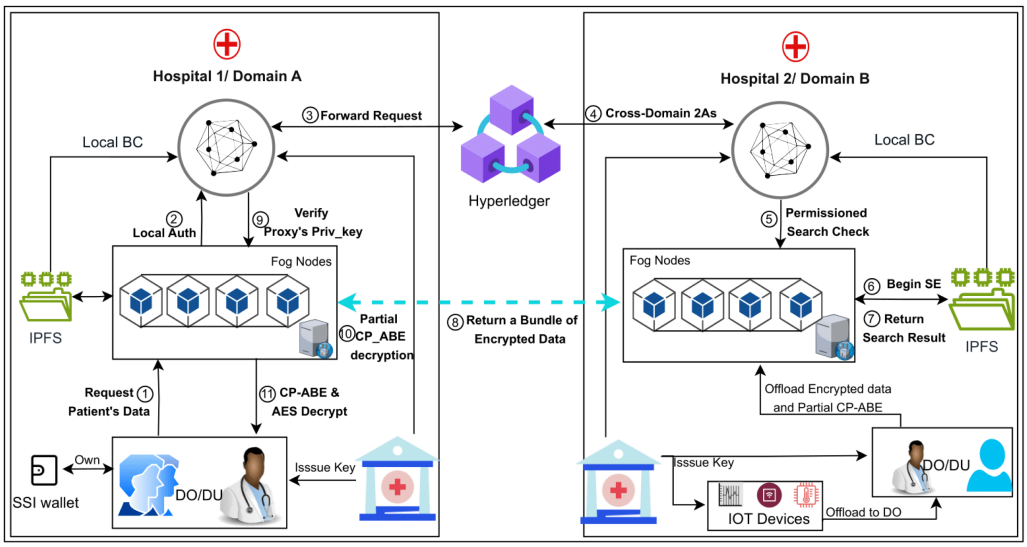

FIGURE 1. SSL-XIoMT System Model.

encrypted data from another domain in the secure-cross-domain manner.

3) Fog Node: contains a proxy server which is a semi-trusted server located in the same network as IFPS responsible for reducing the computational burden for CP-ABE encryption, decryption, and facilitate IoMT data transmission across multiple domains.
4) Hyperledger: is a consortium network blockchain that supports cross-domain authentication and data sharing.
5) Local Blockchain (BC): is the local blockchain in each domain and considered as a member of the whole consortium blockchain network. It has duties for completeness of SSI management and update, flexible query forwarding and responding, local authentication check and validation, and secure key exchange.
6) IPFS: is an interplanetary file system where all encrypted data is stored and managed. Each domain as well as each hospital has an IPFS to keep the encrypted IoMT and encrypted keys. This type of file storage supports cross-domain data sharing and is much more flexible and efficient than normal cloud computing. Its hash function of any file can be shared among its nodes for verification and integrity check.
7) SSI wallet: self-sovereign identity wallet in which data users have freedom of selection to choose any web3 wallet that they are satisfied with.

Also, IoMT data which was executed and transmitted from IoT devices to aggregate terminal and data owners, is also securely conducted such in our previous work [1].

B. CRYPTOGRAPHIC CONSTRUCTION

1) SYSTEM NOTATION

In this section, we demonstrate the system notation which is used throughout this work. Table 1 shows the list of notations used in our scheme.

TABLE 1. Notation

Notation	Description
AA_k	The Trusted Authority k
AE	A set of attributes issued to an Entity E
CT_E	A ciphertext of Entity E
CT_T	Policy Tree of CP-ABE based access policy
nonce	Unique random value
pc	A polynomial coefficient for a universal setup of PLONK
PK_E	(PK_{AA}, PK_I, PK_{DO}, PK_{DU}, PK_{proxy}) Public key issued to an Entity E
RV	Secure random key 256-bit string generated by cryptographically secure pseudorandom number generator
s	Secret value to be hidden and proved by ZKP
SK_E	(SK_{AA}, SK_I, SK_{DO}, SK_{DU}, SK_{proxy}) A secret key issued to an Entity E
SRS	A structured reference string used to set up the cryptographic environment such as ZKPs
v	A public value used in ZKP
VC	Verified credential to be used for SSI verification
π, Π_{ZKPs}	A ZKP proof for SSI Search Privacy; respectively

2) CRYPTOGRAPHIC ALGORITHM

The section presents the details about cryptographic protocols of a secure and self-sovereign identity based blockchain cross-domain authentication and IoMT data sharing. In this chapter, there are six distinct phases which include System Initialization, SSI Generation, ZKP Proof Generation and Verification, Data Encryption, Pre-Computation Before Pairing Operation, and Data Decryption.

Phase1: System Initialization The system initialization phase comprises an algorithm that generates two parts, namely $initializeTrustedAuthority$ and $keyGetFuncGenerator$

Algorithm 1: SSI Generation.

```
function AUTHIDENTITYISSUE(DID, metadata, Hyperledger-network) → VCsigned
    return CSPRNG.SK_I(Req(DID, mdata), TXR)
end function
function CREDENTIALISSUE → Hyper.Sign
    VC_req = Network.Issue(DID, mdata_D, TXR)
    return Hyper.Sign(Sha2Cred(VC_req, DID), IssuePK)
end function
function CREDENTIALVALIDATION → bool
    return crossIdentValidat(VCsigned, DID, Hyper)
end function
function CROSSDOMAINIDENTITYVALIDATION(VCsigned, DID, Hyperledger) → bool
    VC_DU, PK_DU = Hyper.RetrieveCredent(DID)
    return Hyper.VeriCredent(VCsigned, VC_DU, PK_DU)
end function
```

which is executed by the AA. In this phase, we only create and distribute keys and attributes to every DU and DO in the system. The AA executes an algorithm during this phase, namely *keygen()* for AA, DO, DU, and proxy. This algorithm takes as input PK, MK, and AE. The SK of any entities E such as DO, DU, Issuer, proxy, and IPFS is generated by the keygen algorithm using a randomly selected value $r \in \mathbb{Z}_p$, and each attribute $j \in S$ will be represented by randomly selecting $r_j \in \mathbb{Z}_p$.

Phase2: SSI Generation This phase comprises an algorithm called Cross-Domain-Auth-SSI() which is executed via DID and Hyperledger technology. The pseudocode is shown in Algorithm 1.

There are three functions to achieve the cross-domain authentication via SSI management which are $authIdentity\ Issue()$, $credentialIssue()$, and $credentialValidation()$. The $authIdentityIssue()$ generates verifiable credentials (VC-signed) using a decentralized approach by leveraging Multi-Party Computation (MPC). This function processes decentralized identifiers (DID), user metadata, and the Hyperledger network address to create identity credentials that are subsequently updated on the blockchain. Hospitals serve as trusted identity issuers, ensuring that authentication credentials are securely generated and maintained in a tamper-proof manner. The $credentialIssue()$ function then distributes the verifiable credentials to requesters while ensuring that all transactions are transparently recorded on the blockchain, further enhancing security and traceability. Lastly, $credentialValidation()$ is executed as a smart contract to verify whether the requester's identity remains consistent with the blockchain-stored credential, thereby preventing identity fraud. The integration of MPC ensures that sensitive identity computations are distributed across multiple entities, eliminating single points of failure and improving privacy.

Algorithm 2: Optimized ZKP-Proof Generation With MPC and zk-Rollups.

```
function GENERATE_PROOF(s, pc, v, trustiness)
    nonce ← MPC.Rand(RV)
    commitment ← MPC.Hash(polyno(s, pc) + nonce)
    if trustiness is suspicious then
        Store (commitment, nonce) off-chain
    else
        proof ← MPC.Hash(s + polyno(s, pc))
        Store proof off-chain
    end if
    return commitment or proof
end function
function BATCH_PROOFS(pending_proofs)
    return zk-Rollup.Aggregate(pending_proofs)
end function
```

Algorithm 3: Optimized ZKP-Proof Verification With MPC and zk-Rollups.

```
function VERIFY_PROOF(proof, srs, batching)
    if batching then
        batched_proofs ← zk − Rollup.Extract( proof)
        return MPC.VerifyBatch(batched_proofs, srs)
    else
        return MPC.Verify(proof, srs)
    end if
end function
```

Phase3: ZKP Authentication and Authorization Our proposed system integrates Zero-Knowledge Scalable Transparent Argument of Knowledge (ZK-STARK) and PLONK, enhanced with Multi-Party Computation (MPC), to ensure secure, scalable, and efficient user authentication. These cryptographic proofs are generated and verified through smart contracts, as detailed in Algorithms 2 and 3. To optimize blockchain performance, we incorporate off-chain proof precomputation and zk-Rollups, significantly reducing gas costs while preserving security.

Authentication is dynamically adapted based on the user's trust level within the Hyperledger network. For suspicious users without prior recorded interactions, ZK-STARK is employed for proof generation, leveraging its scalability and trustless nature. MPC distributes cryptographic commitments and randomness across multiple entities, ensuring no single party can manipulate the proof data. Proofs are stored off-chain and aggregated via zk-Rollups before submission, minimizing on-chain computation.

For verified users requiring authentication, PLONK is used for both proof generation and verification, benefiting from its efficiency in trusted setups. MPC distributes proof computations across nodes, preventing exposure of complete proof data. To further reduce verification costs, batch verification with zk-Rollups is employed, enabling multiple proofs to be

Algorithm 4: Data Encryption Done by DO.

function POLICYTREE(Node, t, ic) $\rightarrow$ Tree
  CT_M = AES256-Encrypt(M, RV)
  CT_T = policyTree(T.rootNode, t, null)
  return Node
end function
Forward = Pri-K-Enc(CT_M, RV, Node)

Algorithm 5: Data Encryption Done at Fog Node.

function FOG-ENC
  PK-Dec(Forward, PK) $\rightarrow$ CT_M, RV, Node
  CT_T = policyTreeDec(T.rootNode, t, null)
  $hybridPrecomputeResults$ = hybridPreComputePairing(g, h, f, attributes)
  CT_{RV1} = CP-ABE-Encrypt(RV_1, PK,
      PolicyTree, hybridPrecomputeResults)
end function
function HYBRIDPRE(g, h, f, attributes)
  e_{gg}, e_{gh}, e_{gf} = pair(g, g), pair(g, h), pair(g, f)
  att_map = {attr : H.h(attr, $G1$) for attr in atts}
  dynamic_params = dynamicOptimization()
  return (e_{gg}, e_{gh}, e_{gf}, attribute_map, dynamic_params)
end function
function DYNAMICOPTIMIZATION
  dynamicFactor = random()
  return dynamicFactor
end function
function CP-ABE-ENCRYPT(RV_1, PK, PolicyTree)
  e_{gg}, e_{gh}, e_{gf}, attribute_map,
  $C = RV_1 \times e_{gg}^s$
  $C1 = g^s$, $C2 = \{\}$
  for attr in access_structure.attributes **do**
    $C2$[attr] = attribute_map[attr]^s
  end for
  $CT_{RV1} = (C, C1, C2)$
  return CT_{RV1}
end function

validated in a single transaction. This ensures an efficient and scalable proof validation process in a decentralized manner while reducing blockchain transaction costs through off-chain processing and zk-Rollups.

By combining MPC, zk-Rollups, and ZKP protocols, our approach ensures efficient, cost-effective, and privacy-preserving authentication while reducing blockchain overhead, making it suitable for large-scale decentralized identity management.

Phase4: Data Encryption This phase comprises two major parts including (1) $DO - Enc()$ executed by data owners and (2) $proxy - Enc()$ executed by fog node containing proxy server. The first function works on symmetric encryption on EHR using the AES key by DO. This function takes in the message M and RV which returns the Ciphertext of the Message CT_M and the randomly generated number RV.The AES key (RV) is split into two parts: RV_1 and RV_2, where RV_1 is encrypted using CP-ABE and RV_2 is used in ECC-based secure exchange. In the decryption phase, RV_2 is first retrieved and reshuffled with RV_1 to reconstruct the original AES key RV. This ensures that both CP-ABE policy satisfaction and secure exchange are required to recover the symmetric key, thereby strengthening the confidentiality and integrity of the decryption process. DO performs partial CP-ABE-Encryption in which a set of attributes, keyword access, and some security parameters are defined here. We introduced an optimized CP-ABE encryption based on the pre-computing of the mapping attributes and logical gates of the access policy before pairing operation. Algorithm 4 and 5 are the pseudocode representing this data encryption for both data owner and fog.

According to our previous work [1], the technique for partial CP-ABE offloading to fog nodes was explained. After the digital signature is generated using the private key by the DO, a fog node uses the corresponding public key to verify the signature. Next, the proxy in the fog node performs $CP - ABE()$ to encrypt the keywords and finishes the CP-ABE-Enc on the AES key. CP-ABE-ENCRYPT() implements the ciphertext generation for attribute-based access control following the original CP-ABE model by [30]. In Algorithm 5, the variable C_2 is initialized as an empty dictionary to hold attribute-specific ciphertext components used in CP-ABE encryption. For each attribute y defined in the access policy, a corresponding component $C_y = H(y)^s$ is generated, where s is the random session secret and H is a hash function mapping attributes to group elements. This structure enables fine-grained enforcement of access policies during decryption. The initialization $C_2 = \{\}$ follows standard pseudocode notation to denote a key-value mapping between attributes and their encrypted components.

Phase5: Pre-Computation Before Pairing Operation For the pre-computing of the mapping attributes and logical gates of the access control policy before pairing operation, we process attribute layer by layer, starting from the leaf nodes up to the root. This approach ensures that each level is handled independently, reducing the complexity of combining attributes. Also, we separate and prioritize nodes based on their types (And, MofN, OR) and order them as And > MofN > OR. Lastly, we try to balance the tree to minimize the overall depth and evenly distribute the computational load. Fig. 2 presents the comparison of unordered and ordered logical gate of the sample policy tree.

From Fig. 2, it is evident that the traversal paths for the unordered policy tree are as follows:

1) OR $\rightarrow$ AND $\rightarrow$ A1
2) AND $\rightarrow$ MofN $\rightarrow$ A2, A3, A4
3) OR $\rightarrow$ A5 $\rightarrow$ A6

Whereas the traversal paths for the ordered policy tree are:

1) AND $\rightarrow$ MofN $\rightarrow$ A1, A2, A3
2) AND $\rightarrow$ OR $\rightarrow$ A4, A5 $\rightarrow$ A6

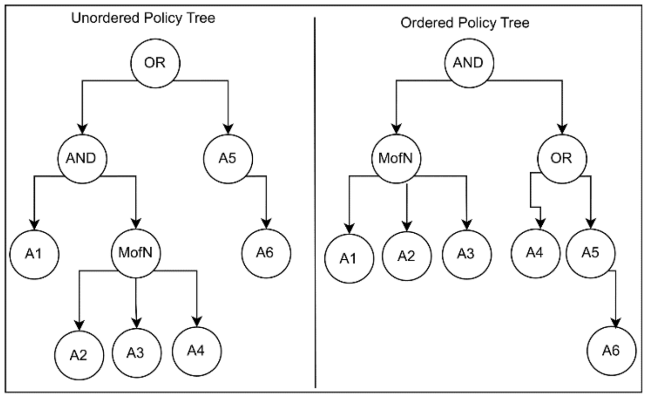

FIGURE 2. Example of Policy Tree Structure.

In the example of an unordered policy tree, when traversing to the root node configured as an OR node, the condition requires that at least one of its child nodes evaluates as true. Specifically, either the left child (an AND node) or the right child (A5) must be true. In the worst-case scenario, if the left child evaluates as false, the system will be forced to evaluate the right child, adding computational overhead. In contrast, within an ordered policy tree where the root is an AND node, if the left child is false, the evaluation halts immediately, avoiding unnecessary checks on the right child. Additionally, in scenarios where an OR operation is evaluated first (as it is less restrictive), the system might traverse more nodes before reaching the more complex M-of-N operation, leading to redundant evaluations and increased computational complexity. Reordering the logical gate hierarchy to prioritize And > MofN > OR can optimize the evaluation process, reducing both pre-computation efforts and the overall complexity of access control policy construction.

Phase6: Data Decryption In this phase, there are five parts to achieve the end target of retrieving EHRs in a secure manner. Those functions are cross-domain authentication via self-sovereign identities, zero-knowledge proof check (authorization), lightweight data exchange with integrity, CP-ABE decryption, and AES-decryption. We break those functions into 3 algorithms. Algorithm 6 shows the authentication for both local and cross-domain, Algorithm 7 shows the lightweight cross-domain data sharing in the cross-domain manner with ECC and Merkle Tree integrity, and Algorithm 8 describes data decryption procedure which is the last process of data retrieval done by the data user.

Starting from Algorithm 6, the process begins with the $localAuth()$ function, which authenticates the Data User (DU) by verifying their decentralized identity (DID) with the local blockchain. The $crossAuthUser()$ function then manages cross-domain authentication by decrypting the user query, completing local authentication, and forwarding the request across the blockchain consortium to the intended destination. The Data User presents verifiable credentials (VC) bundled with zero-knowledge proof (ZKP), which are validated by the verifiers using signed credentials, outputting a verification status. The $Cross-DomainDataSharing()$is executed by a smart contract, checks the authorization of the DU before allowing the $ECC_exchange()$ function to encrypt and securely transfer ciphertexts and keys between domains.

Algorithm 6: Cross-Domain Authentication.

function LOCALAUTH(di, DID, Pri_K_D)
  forward_to_BC(encrypt(DID, Pri_K_D), di)
end function
function CROSSAUTHUSER(DID_u, PK)
  return ident_present(BC.dec(user_query), PK)
end function
function VERIFYCROSSDOMAINAUTHOR($hash_{IPFS}$, $nonce$, VC_{signed}, DID, PP, $Hyper$)
  $merkle_root$ = retrieve_merkle_root(Hyper)
  if not ver_merkle_proof($hash_{IPFS}$, $merkle_root$) **then**
    return False
  end if
  C = con_commit($h_{IPFS} \parallel DID \parallel nonce$)
  $ZKPs$ = hybrid(PP, DID, h_{IPFS}, $nonce$, C, $Hyper$)
  return _ZKP_valid(PP, ZKPs, C, Hyper)
end function

Algorithm 7: Cross-Domain Data Sharing.

function ECC_EXCHANGE
  $time$ = generate_time()
  SK_{OA} = key_gen($RV \oplus SK_{OB}$)
  CT_{AU} = encrypt_data_retrieval(CT_M, CT_{RV}, SK_{OA}, CT_{DU}, $time$, Proxy_A)
  return CT_{AU}
end function
function VERIFVZKP(PP, Π_{ZKP}, C)
  $merkle_root$ = retrieve_merkle_root(Hyperledger)
  if not verify_merkle_proof(C, merkle_root) **then**
    return False
  end if
  return check_ZKP_validity(PP, Π_{ZKP}, C)
end function

Algorithm 8: Data Decryption.

function DUDEC(RV, CT_M)
  Divide(R') $\rightarrow R_2 \parallel R_1$
  ReShuffle($R_2 \parallel R_1$) $\rightarrow (R_1 \parallel R_2) = R$
  Dec(Enc(SK_{proxy}, R)) $\rightarrow SK_{\text{proxy,A}}$
  Dec(CT_{AU}, $SK_{\text{proxy,A}}$)
  return CT_L, CT_{AU}, $SK_{\text{DO-DU}}$, $time$
  if $time$.valid == "True" **then**
    RV_1 = CP_ABE_Dec(CT_{RV1}, $SK_{\text{DO-DU}}$)
    $RV = RV_1 + RV_2$
    M = AES_Dec(CT_M, RV)
    return M
  end if
end function

The proxy in domain B encrypts the data, keys, and integrity checks using the public key of the proxy in domain A, creating a secure transmission CT_{DU} to domain A. The $Verifvzkp()$ function verifies authorization claims using Merkle Tree-based integrity checks recorded on the blockchain, ensuring secure and trustworthy data transmission. This process does not rely on ECC-based cryptography, as Merkle Tree verification more effectively supports multi-component integrity validation without the computational overhead of ECC. If verified, the proxy in domain A decrypts the received data and checks the validity of the time. If valid, it proceeds with the decryption of CP-ABE using RV_2 which is bundle from BC associated to index of CT_{RV1} during encryption stage. Lastly, DU can perform AES and access the data (M).

IV. SECURITY ANALYSIS

A. SECURITY MODEL

In this part, we highlight the key considerations to make sure that our system guarantees a high security level. To evaluate specific algorithms of our system, we break the security model into 3 parts such as ZKP protocols, Merkle integrity, and Pre-Computation before pairing operation for CP-ABE. We have made several assumptions as follows: 1. No PPT adversary can break the soundness of ZK-STARK and Plonk with non-negligible advantage. 2. No PPT adversary can break the zero-knowledge property of ZK-STARK and PLONK with non-negligible advantage. 3. No PPT adversary can break the integrity of a Merkle tree with non-negligible advantage. 4. No PPT adversary can forge a valid proof for a non-existent leaf in a Merkle tree with non-negligible advantage. 5. No PPT adversary can break the strength of CP-ABE by the use of pre-computed logical operations before the pairing operation.

B. SECURITY ANALYSIS

Proof 1: Let A be a PPT adversary attempting to produce a proof π for a false statement. We define the soundness experiment $\mathrm{ExpA}_{\mathrm{sound}}(\lambda)$ where λ is the security parameter.

1. *Initialization:* The challenger C runs the setup algorithm to generate the common reference string $\mathrm{crs} \leftarrow \mathrm{Setup}(\lambda)$ for ZK-STARK and structured reference string (SRS) $\mathrm{srs} \leftarrow \mathrm{Setup}(\lambda)$ for PLONK.

2. *Adversary's Move:* A outputs a proof π, π' and a statement κ, κ'.

3. *Verification:* C checks if π, π' is a valid proof for κ, κ' using the verification algorithm $\mathrm{Verify}(\mathrm{crs}, \kappa, \pi)$ for ZK-STARK and $\mathrm{Verify}(\mathrm{srs}, \kappa, \pi)$ for PLONK. If Verify returns 1 (accepts) for a false statement κ, κ', A wins. The probability that A wins, denoted as $\mathrm{AdvA}_{\mathrm{Sound}}(\lambda)$, is negligible, assuming the soundness of ZK-STARK, which is based on the hardness of underlying cryptographic assumptions such as the difficulty of solving certain classes of polynomials over finite fields.

Therefore, $\mathrm{AdvA}_{\mathrm{Sound}}(\lambda) \leq \mathrm{negl}(\lambda)$, proving the soundness of ZK-STARK. Assuming the cryptographic assumptions underlying PLONK, such as the knowledge of exponent assumptions, the probability that A wins is negligible:

$\mathrm{AdvA}_{\mathrm{Sound}}(\lambda) \leq \mathrm{negl}(\lambda)$.

Proof 2: Let A be a PPT adversary. Consider the zero-knowledge experiment $\mathrm{ExpA}_{\mathrm{ZK}}(\lambda)$:

1. *Setup:* The simulator S generates crs, srs and a trapdoor T, T' for ZK-STARK and PLONK, respectively.

2. *Simulation:* S simulates a proof π_{proof} for any given statement κ without knowing the witness. Using T', S generates a proof π'_{proof} for any statement κ'.

3. *Indistinguishability:* The distributions of π_{real} and π_{sim} must be computationally indistinguishable. As well as π'_{sim} must be indistinguishable from π'_{real}. For any adversary A, there exists a simulator S such that:

$|\Pr[A(\pi_{\mathrm{real}}) = 1] - \Pr[A(\pi_{\mathrm{sim}}) = 1]| \leq \mathrm{negl}(\lambda)$

$|\Pr[A(\pi'_{\mathrm{real}}) = 1] - \Pr[A(\pi'_{\mathrm{sim}}) = 1]| \leq \mathrm{negl}(\lambda)$

This implies that A cannot distinguish between a real proof and a simulated proof, ensuring the zero-knowledge property of ZK-STARK and PLONK.

Proof 3: Let A be a PPT adversary attempting to forge a valid authentication path for a falsified leaf in a Merkle tree. We define the integrity experiment $\mathrm{ExpA}_{\mathrm{int}}(\lambda)$ where λ is the security parameter.

1. *Initialization:* The challenger C generates a Merkle tree M with leaves $L_1, L_2, \ldots, L_n$. Each leaf L_i is associated with a data block D_i. The tree is constructed using a cryptographic hash function H.

2. *Adversary's Move:* A outputs a leaf L^*, a data block D^*, and an authentication path Path^*.

3. *Verification:* C checks if the authentication path Path^* is valid for L^* in the Merkle tree M. The experiment is defined as follows:

$$\mathrm{ExpA}_{\mathrm{int}}(\lambda) = \begin{cases} 1 & \text{if } \mathrm{Verify}(M, L^*, D^*, \mathrm{Path}^*) = 1 \text{ and} \\ & \quad L^* \neq L_i \text{ for any } i \\ 0 & \text{otherwise} \end{cases}$$

If Verify returns 1 for L^* and Path^* is not associated with any L_i, A wins. The integrity of the Merkle tree relies on the collision resistance and pre-image resistance properties of the hash function H. Given the collision resistance of H, the probability that A can find two distinct inputs x and y such that $H(x) = H(y)$ is negligible:

$\mathrm{AdvA}_{\mathrm{int}}(\lambda) \leq \mathrm{negl}(\lambda)$.

Proof 4: Let A be a PPT adversary attempting to forge a proof for a non-existent leaf L^*. Consider the proof-forgery experiment $\mathrm{ExpA}_{\mathrm{pf\text{-}forge}}(\lambda)$:

1. *Setup:* The challenger C generates a Merkle tree M with root R.

2. *Adversary's Move:* A outputs a leaf L^*, a data block D^*, and a proof Path^*.

3. *Verification:* C checks if Path^* is a valid proof for L^* under the root R. The experiment is defined as follows:

$$\mathrm{ExpA}_{\mathrm{pf\text{-}forge}}(\lambda) = \begin{cases} 1 & \text{if } \mathrm{Verify}(M, L^*, D^*, \mathrm{Path}^*) = 1 \text{ and} \\ & \quad L^* \notin M \\ 0 & \text{otherwise} \end{cases}$$

If Verify returns 1 for L^* and L^* is not in M, A wins. Given the collision resistance of H and the structure of the Merkle tree, the probability that A can forge a valid proof for a non-existent leaf is negligible:

$\mathrm{ExpA}_{\text{pf-forge}}(\lambda) \leq \mathrm{negl}(\lambda)$.

Proof 5: The introduction of pre-computed logical operations before the pairing operation does not compromise the security of the CP-ABE scheme.

1. *Initialization and Pre-Computation:* The Trusted Authority generates a key pair, and the simulator generates additional key pairs. Logical operations are pre-computed before the pairing operation to optimize the construction of access control policies.

2. *Phase 1:*

- Adversary A sends a set of attributes S to Challenger C.
- Challenger C provides the secret key SK corresponding to the attributes in S to Adversary A.

3. *Challenge:*

- Adversary A sends two messages m_1 and m_2 of equal length to the simulator.
- A fair binary coin β is flipped by the simulator, and the encryption of m_β is returned.
- The ciphertext CT is computed using pre-computed values as:

$$CT = \big(T, C'' = m_\beta Z, CT_T = h^s, \forall y \in Y : C_y = g^{q_y(0)},$$
$$C'_y = H(\mathrm{att}(y))^{q_y(0)}\big)$$

  where CT_T refers to attribute dependent value and s is attributes associated with encryption and a subset of S.
- If $\mu = 0$, $z = e(g, g)^{\alpha s}$. Hence, the ciphertext is a valid random encryption of m_β.
- If $\mu = 1$, $z = e(g, g)^z$. Consequently, $C'' = m_\beta e(g, g)^z$, making C a random element of G_1 from Adversary A's perspective, thus revealing no information about m_β.

4. *Phase II:* The simulator repeats the process from Phase I.

5. *Guess:* Adversary A sends a guess β' of β. The advantage of Adversary A is computed as:

$$\text{Advantage} = \Pr[\beta = \beta'] - \frac{1}{2}$$

Conclusion: The pre-computed logical operations do not introduce any vulnerabilities, and our scheme remains secure under the same assumptions as the original CP-ABE scheme.

V. EVALUATION

A. FUNCTIONALITY COMPARISON

This section presents a comparison of the features of our proposed system and related works including [27], [31], [32], [33], and [34]. Table 2 presents a comparison between our scheme and these related works across five distinct functions which integrate identity-based management or SSI within the cross-domain authentication manner.

The comparative analysis of identity-based management systems for cross-domain authentication reveals limitations in existing schemes. While Scheme [27] leverages CP-ABE and proxies, it lacks ZKP flexibility and post-quantum resistance. Scheme [28] uses Hyperledger and Ethereum for anonymous authentication but also misses post-quantum resistance and pre-compute pairing. Scheme [29] integrates DABE and ZKPs but is not optimized for lightweight IoMT transmission or post-quantum security. Scheme [30] offers post-quantum-resistant authentication with ECC but lacks pre-compute pairing access control, and Scheme [31] combines SSI with ZKPs but omits lightweight IoMT support and post-quantum features. In contrast, our proposed system integrates SSI, ZK-STARK, PLONK, and CP-ABE, providing flexible ZKPs, post-quantum resistance, pre-compute pairing, lightweight IoMT transmission, and Merkle integrity checks, offering superior security and efficiency for cross-domain data sharing.

TABLE 2. Functionality Comparison

Scheme	F1	F2	F3	F4	F5	F6
[27]	X	✓	X	X	X	X
[31]	X	✓	✓	X	X	X
[32]	X	X	X	X	X	X
[33]	X	X	X	X	X	✓
[34]	✓	X	X	X	X	✓
Ours	✓	✓	✓	✓	✓	✓

Note: F1: Flexible ZKPs, F2: CP-ABE offloading Enc/Dec, F3: Lightweight IoMT data transmission, F4: Post-Quantum Resistance, F5: Pre-Compute-Pairing Access Control Policy, F6: Integrity Check.

B. PERFORMANCE ANALYSIS

In this section, we conducted experiments to compare the processing time for cross-domain authentication and authorization of our proposed system and related schemes [28], [29], and [31]. We also measured the performance of encryption, decryption, re-encryption, data transmission, and integrity checking. Finally we conducted the simulation to measure system throughput.

In this section, we evaluate and compare the performance of our proposed system against related schemes [28], [29], and [31] in terms of cross-domain authentication and authorization. We also assess encryption, decryption, re-encryption, secure data transmission, integrity verification, and overall system throughput under varying loads.

All experiments were performed on a server running Ubuntu 20.04, equipped with an Intel Xeon E-2336 CPU @ 2.9 GHz and 16 GB RAM. The implementation was carried out in Python, along with cryptographic libraries such as PyCryptodome [35], Charm-Crypto [36], and OpenSSL [37]. For attribute-based encryption and pairing operations, we utilized Docker-CP-ABE [38] and pairing-operation [39]. Zero-knowledge proofs were implemented using Rust libraries, including libstark [40] and bellman [41]. Cross-domain authentication and identity handling were managed via Hyperledger Aries Cloud Agent Python (ACA-Py) [42].

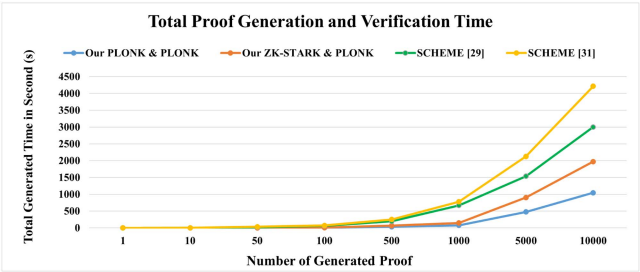

FIGURE 3. Total Proof Gen. and Ver. Time.

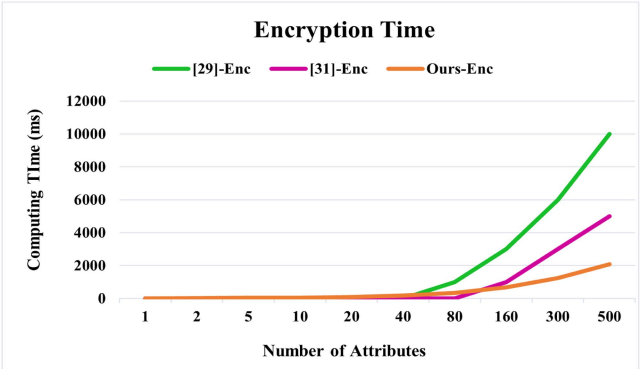

FIGURE 4. Attribute-Based Encryption Comparison.

• TOTAL PROOF GENERATION AND VERIFICATION TIME

To evaluate the performance of proof generation and verification, we simulated up to 10,000 ZKPs using Python integrated with Rust and Java libraries. Fig. 3 compares our proposed scheme against Schemes [29] and [31], incorporating both proof creation and trusted setup times. As shown in Fig. 3, when the number of proofs was ≤ 1000, Scheme [31] (using standard ZK-SNARK) incurred the highest delay (522.3 seconds), while Scheme [29] and our system performed faster—at approximately 109.7 seconds and 69.4–76.8 seconds, respectively. As the volume increased to 5,000–10,000 proofs, Scheme [31] outpaced Scheme [29] due to its optimized verification path, but still lagged behind our design in total efficiency.

Our hybrid ZKP framework leverages PLONK for fast verification and ZK-STARK for trustless scalability. In contrast, Scheme [29] uses at-zkp, which offers flexibility but suffers from repeated trusted setup overheads. This demonstrates that our integration of PLONK and simplified ZK-STARK provides better scalability, lower setup cost, and stronger suitability for cross-domain SSI scenarios.

• ENCRYPTION AND DECRYPTION TIME

We evaluated the encryption and decryption performance of our scheme against Schemes [29] and [31] using CP-ABE. Each test used a constant data size of 10,000 KB, while the number of attributes was varied to assess scalability. Figs. 4 and 5 illustrate the overall computation cost for encryption and decryption, respectively.

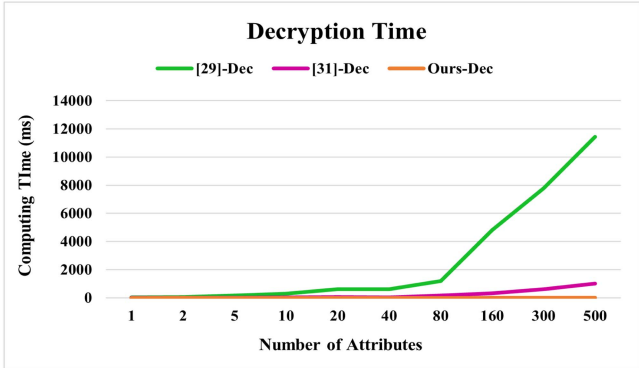

FIGURE 5. Decryption Time Comparison.

TABLE 3. Data Transmission Across Domains

Scheme	Setup Time	PKI / ECC Encryption	IPFS Verification	Total Time
[28]	0.00260s	0.001034s	2.55107s	2.56s
[29]	0.0285s	0.008s	0.00071s	0.03721s
Ours	0.00236s	0.00724s	0.0005s	0.01018s

As shown in Fig. 4, our system consistently achieved the fastest encryption time across all attribute sizes. Scheme [29], which employs Decentralized Attribute-Based Encryption (DABE), exhibited the highest computation cost, while Scheme [31] performed moderately well. For attribute counts below 160, Scheme [31] and our system show comparable performance. However, as the number of attributes increased—particularly up to 300—the performance gap widened, with our system maintaining significantly lower encryption latency. This improvement is due to our use of pre-computation for attribute mapping and logical gate optimization before the pairing operations.

Fig. 5 shows a similar trend for decryption. Our system outperformed both baselines, followed by Scheme [31], with Scheme [29] again showing the slowest performance. The efficient structure of our access control tree and the lightweight pairing strategy contribute to the improved decryption speed.

• SECURE CROSS-DOMAIN DATA TRANSMISSION

We measured the total time required to securely transmit encrypted data across domains, focusing on three key functions: setup (e.g., PKI key generation and parameter initialization), ECC-based encryption, and IPFS-based integrity verification. Since only Schemes [28] and [29] implement similar cross-domain data transmission mechanisms, we compared them with our proposed system. Table 3 summarizes the time taken for each step.

As shown in Table 3, our system consistently achieved the lowest overall latency. The setup includes cryptographic parameter generation and randomness initialization. ECC-based encryption secures the payload under public/private key pairs, while integrity verification uses IPFS hashes distributed among storage nodes. While Scheme [29] emphasizes stronger security characteristics, it exhibits higher transmission latency than our system. Scheme [28] achieves a more

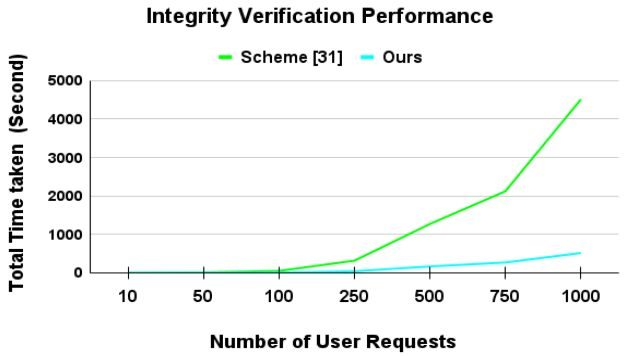

FIGURE 6. Integrity Verification Performance.

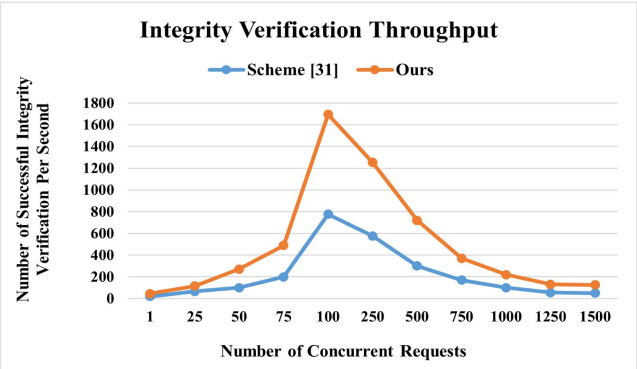

FIGURE 7. Integrity Verification Throughput.

balanced trade-off but underperforms in both efficiency and cryptographic assurance.

All schemes completed the end-to-end process in under 0.1 seconds, but our system showed the best overall performance with faster execution across all steps.

• INTEGRITY VERIFICATION PERFORMANCE AND THROUGHPUT

We compared the integrity verification performance and throughput of our proposed system against Scheme [31]. Our system combines Merkle Tree verification with IPFS hashing to ensure verifiable integrity, whereas Scheme [31] relies mostly on blockchain immutability and policy-level integrity through smart contracts. Figs. 6 and 7 illustrate the strength of our combined confidentiality and integrity mechanism in terms of performance and throughput.

As shown in Fig. 6, our system outperforms Scheme [31] when the number of requests exceeds 500, demonstrating better scalability for high-volume environments. Although Scheme [31] shows slightly lower latency under minimal load (< 50), our Merkle-based design handles larger loads more effectively.

Fig. 7 illustrates throughput in terms of successful integrity verifications per second. Initially, Scheme [31] had better performance, but our system surpassed it as concurrent requests exceeded 50, peaking at 918 verifications per second (100–150 users), compared to Scheme [31]'s maximum of 777. Beyond 750 concurrent users, both systems experienced reduced throughput, though our system consistently maintained a slight edge. Overall, both schemes ensure both authenticity and granular integrity for cross-domain data sharing beyond what is provided by ECC encryption alone or by blockchain logging.

VI. CONCLUSION

SSL-XIoMT offers a secure and scalable solution for cross-domain IoMT data sharing by combining SSI, ZK-STARK, PLONK, MPC, and blockchain to enable efficient, privacy-preserving authentication. It reduces computational overhead through off-chain proof generation and zk-Rollups, lowering gas costs and improving blockchain scalability. The system enhances CP-ABE by precomputing access policies and supports ECC-based keys, Merkle tree verification, and Hyperledger integration for robust identity management. Experiments confirm its efficiency and security. Future work includes adding searchable encryption for private queries and using federated learning for collaborative healthcare analytics.

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